# Indefinite-signature metrics and Levi-Civita Christoffel uniqueness: a Lean library closing the curvature gap above the Bonn covariant-derivative API


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The Massot-Rothgang-Macbeth covariant-derivative API (`Mathlib.Geometry.Manifold.VectorBundle.CovariantDerivati` landed in Mathlib in April 2026, but Mathlib still has no Riemann, Ricci, or scalar curvature on a connection, and no indefinite-signature metric. Both gaps are out of HALF-ERC scope. We close them at the algebraic-precedent level with seven Lean modules: `LorentzianMetric.lean` ships an indefinite-signature metric typeclass with the Minkowski $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ witness and a Riemannian-Lorentzian signature falsifier; `RiemannianConnection.lean` ships the Christoffel-quadratic part of the Riemann tensor with an algebraic antisymmetry-in-$(\mu, \nu)$ proof, the Koszul-formula closed form for Levi-Civita Christoffels, and a torsion-freeness theorem under metric-partial symmetry (the substantive Levi-Civita uniqueness theorem — every torsion-free metric-compatible connection equals the Koszul Christoffel — needs the bundle-level Koszul-bilinear-form argument over arbitrary smooth vector fields and is deferred to the upstream-port wave so it lands as a substantive theorem rather than a predicate-level trivial-trans on a $\Gamma =$ `christoffelKoszul` def-as-equality); `RiemannCoordinate.lean` ships the full coordinate Riemann tensor including the linear-in-$\partial\Gamma$ piece and the algebraic first Bianchi identity for the full Riemann under torsion-freeness on both the Christoffel and its partial-derivative array; and `RiemannDifferentialBianchi.lean` ships the second-derivative carrier $\partial^2\Gamma$ with a Schwarz-symmetry hypothesis, the covariant-derivative operator $\nabla_\lambda R^\rho{}_{\sigma\mu\nu}$ on a generic Riemann tensor, the linear-piece differential second Bianchi identity under Schwarz alone, and the Group C+D quadratic cyclic cancellation under torsion-freeness plus `AntisymLastTwo`; `BundleRiemannAux.lean` opens the Bonn-API bundle scope with a bare-function bundle Riemann curvature $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$ consuming `IsCovariantDerivativeOn` and `VectorField.mlieBracket`, with the wave-headlines $R(X, X)Z = 0$ (from `mlieBracket_self`) and $R(X, Y)Z = -R(Y, X)Z$ (from `mlieBracket_swap_apply` plus continuous-linearity of the $\mathrm{cov}\,\sigma\,x$ slot), plus a commuting-vector-fields specialisation and zero-degeneracies for the section and the X-slot on Bonn's `IsCovariantDerivativeOn.zero`; `BundleRiemann.lean` ships pointwise cyclic Jacobi for mlieBracket (Mathlib has only the Leibniz form), the abstract bundle-level first Bianchi identity for $V := \mathrm{TangentSpace}\, I$ under torsion-freeness, and root-namespace `IsCovariantDerivativeOn.neg / .sub` Mathlib-style helpers that discharge the abstract section-additivity hypothesis when the connection comes from Bonn; and `LeviCivita.lean` ships the substantive bundle-level Levi-Civita uniqueness theorem via the Wald §3.1 / Kobayashi-Nomizu IV.2.2 Koszul-bilinear-form derivation $2\, g(\nabla_X Y, Z) = Xg(Y, Z) + Y g(Z, X) - Zg(X, Y) + g([X, Y], Z) - g([X, Z], Y) - g([Y, Z], X)$ from IsCovariantDerivativeOn + torsion-free + metric-symmetric + metric-compatible (plus $\mathrm{char}(\mathbb{K}) \neq 2$ for the 2-cancellation in the composition uniqueness theorem), plus the algebraic uniqueness kernel routing via bilinearity non-degeneracy and `FiberBundle.extend`. To our knowledge no proof assistant has previously formalized cyclic Jacobi for the manifold Lie bracket (Mathlib4 ships only the Leibniz form), bundle-level first Bianchi torsion-free for $V := \mathrm{TangentSpace}\, I$, or substantive bundle-level Levi-Civita uniqueness via the Koszul-bilinear-form derivation; per-system search of Mathlib4, Mathcomp-Analysis, HOL Light Multivariate, HOL4, Isabelle/AFP, Mizar MML, and Agda agda-categories / agda-unimath (via web search + Phase 6f deep-research audit §3E) found no positive prior art through 2026-04-30, but the survey is not exhaustive and a per-system appendix is queued for Wave 8 closure. The library follows Mathlib upstream conventions for an eventual upstream port; `BundleRiemannAux.lean` + `BundleRiemann.lean` + `LeviCivita.lean` together ship the bundle-Riemann + bundle-Levi-Civita backbone and are the deliverable nucleus of a Mathlib-shaped pull request.


## I. INTRODUCTION

The 2025 Massot-Rothgang-Macbeth covariant-derivative library landed in Mathlib in early 2026 (commit `81a5d257`, files dated April 13, 2026), supplying `IsCovariantDerivativeOn`, `CovariantDerivative`, and the torsion-tensor companion [1]. The HALF-ERC project's published scope explicitly excludes Riemann, Ricci, and scalar curvature on a connection; on the metric side, Mathlib's `IsRiemannianManifold` (Gouëzel) and `IsContinuousRiemannianBundle` are positive-definite and offer no Lorentzian analog.

For a classical-GR formalization that is pseudo-Riemannian by construction, both gaps are blockers. The author's project SK-EFT-Hawking has formalized the algebraic Riemann tensor (`SKEFTHawking.Curvature`, 2026-04-29 ship), the Einstein tensor (`EinsteinTensor`), exact solutions (Minkowski, de Sitter, anti-de Sitter, Schwarzschild via Kerr-Schild form), the ADM 3+1 decomposition, the tetrad formalism, causal structure, the Penrose and Hawking-Penrose singularity hypothesis bundles, the classical area theorem in its monotone-mass form, and the no-hair-theorem Kerr-family substrate [3, 4]. Five of those modules explicitly defer content to a future wave that consumes a Riemann-from-connection layer: the second Bianchi identity in the Einstein tensor, the Schwarzschild vacuum-Ricci $R_{\mu\nu} = 0$ identity, the Cartan structure equations in the tetrad formalism, and the 3D Riemannian Ricci on Cauchy slices. None can be discharged without the Mathlib gap above closed.

This paper documents seven new Lean modules that close both gaps at the algebraic-precedent level and open the bundle scope above the Bonn API. The treatment follows the project's algebraic-precedent conventions established in the 6f and 6g programmes: metric matrices are $\mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$, Christoffel symbols are $\mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$ (one upper, two lower), and partial derivatives of the metric are passed as explicit input arrays $\partial g \in \mathrm{Fin}\,4 \to \mathrm{MetricMatrix}$ (eventually replaced by the manifold `mfderiv` in the upstream port, but with the algebraic identities the same). At this scope the library ships fifty-two theorems and lemmas across seven modules, zero `sorry`, zero new project axioms; all load-bearing theorems verify against `propext`, `Classical.choice`, `Quot.sound` only. The post-wave-strengthening cut history across Sessions 1-5b is documented in Section XII A (the W7+W8(S1-3) reflexive strengthening pass cut 2 retroactive theorems on 2026-04-30; a same-day ruthless audit on Sessions 4a/4b/5/5b cut 4 additional section-level `funext`/`FiberBundle.extend` lifts as P3 plumbing without consumers — see Section XII B). All cuts retired predicate-level trivial-trans / corollary patterns that had no downstream consumers and whose substantive content lives elsewhere in the library.

As of W8 Sessions 5 + 5b (Section X) we now claim the substantive bundle-level Levi-Civita uniqueness theorem itself: every torsion-free metric-compatible covariant derivative on the tangent bundle agrees pointwise with every other such connection, under IsCovariantDerivativeOn + torsion-free + metric-symmetric + metric-compatible and $\mathrm{char}(\mathbb{K}) \neq 2$. The proof routes via the Wald §3.1 / Kobayashi-Nomizu IV.2.2 Koszul-bilinear-form derivation (Section X.3), the algebraic uniqueness kernel (Section X.1), and a 2-cancellation step. We also retain the closed-form Levi-Civita Christoffel via the Koszul formula at coordinate scope, with torsion-freeness under metric-partial symmetry. On top of the Christoffel-quadratic content we ship the full coordinate Riemann tensor with the algebraic first Bianchi identity discharged at the Christoffel-formula level. W7's commutator-only quadratic part trivially satisfies the cyclic Bianchi sum by structural ring discharge (the bundled-covariant-derivative formulation $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$ makes this automatic); but at the algebraic Christoffel-formula level the cyclic-sum cancellation requires explicit twelve-rewrite torsion-free symmetry chains that we discharge in this paper.

## II. LORENTZIAN METRIC

### A. Algebraic-precedent signature predicate

A metric matrix $g : \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$ has *Lorentzian signature* if $g_{00} = -1$, $g_{ii} = +1$ for $i \in \{1, 2, 3\}$, and $g_{\mu\nu} = 0$ for $\mu \neq \nu$. The canonical Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ satisfies this predicate; the proof discharges directly via three existing lemmas in `LinearizedEFE` ($\eta$`_zero_zero`, $\eta$`_spatial_diag`, $\eta$`_off_diag`).

**Theorem 1** (Riemannian-Lorentzian falsifier)**.** *No metric matrix simultaneously satisfies* `IsRiemannianSignature` *and* `IsLorentzianSignature`*. Evaluation at* $(0, 0)$ *gives* $+1 = -1$*, contradiction.*

This signature-distinguishing falsifier is not vacuous: both predicates are witnessed (Riemannian by the Euclidean identity, Lorentzian by $\eta$), and the falsifier consumes both bundles to extract the contradicting diagonal entry.

### B. Bundle structure

`IsLorentzianMetric` bundles symmetry, the signature predicate, and (by consequence of the diagonal product) non-degeneracy: $\det g = -1$ for the strict-diagonal Lorentzian, and $g_{00} = -1 \neq 0$. The non-vacuity witness is again the Minkowski metric, which inherits symmetry from the existing $\eta$`_symm` theorem.

### C. Mathlib-style typeclass shape

A family $g : B \to \mathrm{MetricMatrix}$ over a base type $B$ is *continuously Lorentzian* (notation: `IsContinuousLorentzianFamily`) iff every fiber metric is `IsLorentzianMetric`. The constant-Minkowski family $b \mapsto \eta$ over any $B$ is continuously Lorentzian. This is the algebraic-precedent shadow of the eventual Mathlib-PR target `IsContinuousLorentzianBundle F E`, which mirrors Gouëzel's `IsContinuousRiemannianBundle F E` on a vector bundle with `[FiberBundle F E]` and `[VectorBundle ℝ F E]` plus a continuity-of-section requirement [2].

## III. CHRISTOFFEL-QUADRATIC RIEMANN

The Riemann curvature tensor in Christoffel-symbol form is

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}. \tag{1}$$

Without modeling $\partial_\mu \Gamma$ at the algebraic-precedent scope, we ship the quadratic part as a substantive object:

$$(\Gamma\Gamma)^\rho{}_{\sigma\mu\nu} := \sum_\lambda \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \sum_\lambda \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}. \tag{2}$$

**Theorem 2** (Christoffel-quadratic antisymmetry in $(\mu, \nu)$)**.** *For any* $\Gamma : \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$, $(\Gamma\Gamma)^\rho{}_{\sigma\mu\nu} = -(\Gamma\Gamma)^\rho{}_{\sigma\nu\mu}$.

The proof is a single `ring` discharge after unfolding: the defining expression $A - B$ swaps to $B - A$ on $(\mu, \nu)$ exchange. This connects `riemannQuadraticTerm` to `Curvature.AntisymLastTwo` (SK-EFT-Hawking 6f.1) via a definitional cross-bridge: `leviCivitaRiemannQuadratic_AntisymLastTwo` consumes `riemannQuadraticTerm_AntisymLastTwo`.

## IV. KOSZUL FORMULA

The Koszul-formula closed form for the Levi-Civita Christoffel symbols on a metric $g$ with inverse $g^{-1}$ and partial derivatives $\partial g$ is

$$\Gamma^\rho{}_{\mu\nu} = \tfrac{1}{2} \sum_\sigma g^{\rho\sigma} \big(\partial g_\mu g_{\sigma\nu} + \partial g_\nu g_{\sigma\mu} - \partial g_\sigma g_{\mu\nu}\big). \tag{3}$$

We encode (3) as `christoffelKoszul gInv dg : Christoffel`, a noncomputable function (the 1/2 factor depends on the noncomputable `Real.instDivInvMonoid`). The substantive content is in two theorems:

**Theorem 3** (Koszul Christoffel is torsion-free)**.** *Under* `MetricPartialSymmetric dg` *($\partial g_\mu g_{\sigma\nu} = \partial g_\mu g_{\nu\sigma}$ for all $\mu, \sigma, \nu$), the Koszul Christoffel satisfies $\Gamma^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu}$.*

The proof is a four-term `linear_combination` discharge, weighting the four $\sigma$-instances of metric-partial symmetry by the inverse-metric components $g^{\rho\sigma}/2$. The hypothesis is genuinely load-bearing: a non-symmetric $\partial g$ would yield $\Gamma^\rho{}_{\mu\nu} \neq \Gamma^\rho{}_{\nu\mu}$ in general, encoding torsion in the connection.

*a. Levi-Civita uniqueness — strategic deferral.* The textbook fundamental theorem of (pseudo-)Riemannian geometry [6, 7] reads: every torsion-free metric-compatible connection on $(M, g)$ equals the Koszul-formula Christoffel (3). At our algebraic-precedent scope a tempting but *trivial* encoding ships a predicate `IsLeviCivitaForKoszul` $\Gamma\, g^{-1}\, \partial g := \Gamma =$ `christoffelKoszul` $g^{-1}\, \partial g$ and proves "uniqueness" by definitional transitivity ($\Gamma = K \wedge \Gamma' = K \Rightarrow \Gamma = \Gamma'$ via `rw`). That theorem reduces to transitivity-of-equality through unfolding, with the substantive content living entirely in `christoffelKoszul`'s closed form rather than in any uniqueness statement. Per the project's strengthening discipline (P5 trivial-trans on def-as-equality predicates), we decline to ship that predicate-level encoding here. The substantive Levi-Civita uniqueness — every torsion-free metric-compatible connection equals the Koszul Christoffel on smooth vector fields, by the Koszul-bilinear-form argument [7] — needs the bundle-level `IsCovariantDerivativeOn` API over arbitrary smooth sections, and SHIPPED in Wave 8 Sessions 5 + 5b (§X: `leviCivita_pointwise_unique_of_koszul` algebraic uniqueness kernel + substantive `IsLeviCivita` predicate + `koszul_identity_of_isLeviCivita` Wald §3.1 derivation + `leviCivita_unique_of_isLeviCivita` composition theorem under char($\mathbb{K}$) $\neq$ 2). At the algebraic-precedent scope of this section, Theorem 3 discharges the load-bearing half (the Koszul Christoffel *is* torsion-free under metric-partial symmetry); the converse half (every torsion-free metric-compatible $\Gamma$ *is* the Koszul Christoffel) is now formalised at the bundle scope in §X.

## V. CROSS-BRIDGE: FLAT SPACE HAS ZERO QUADRATIC CURVATURE

The canonical sanity check for any classical-GR formalization is that flat spacetime has zero Riemann curvature. At our scope:

**Theorem 4** (Constant-metric vanishing curvature)**.** *If* `MetricPartialZero dg` *(all components vanish), then the Levi-Civita quadratic Riemann* `leviCivitaRiemannQuadratic gInv dg` *equals the zero tensor.*

The proof composes two intermediate results: (a) the Koszul Christoffel of a constant metric vanishes, by direct algebraic discharge under the zero hypothesis applied at all twelve $(\sigma, \mu, \nu)$ arguments; (b) the quadratic Riemann of the zero Christoffel vanishes, by definitional unfolding plus `ring`.

The result instantiates as expected on the constant Minkowski background $\eta$: $\partial g \eta \equiv 0$ at all chart points, so the algebraic-precedent vanishing-curvature theorem applies, and the Christoffel-quadratic Riemann of flat Lorentzian space is zero.

## VI. FULL COORDINATE RIEMANN TENSOR AND ALGEBRAIC FIRST BIANCHI

The Christoffel-quadratic content shipped in §III–§V is only the second half of the coordinate Riemann formula (1). The first half — the linear-in-$\partial\Gamma$ piece — carries the load-bearing content for the algebraic first Bianchi identity. `RiemannCoordinate.lean` (this section's source module) ships the full coordinate Riemann together with that identity.

### A. Linear-in-$\partial\Gamma$ piece

We model the partial-derivative array of the Christoffel symbol as $d\Gamma : \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$, with the convention $d\Gamma\,\mu\,\rho\,\nu\,\sigma = \partial_\mu \Gamma^\rho{}_{\nu\sigma}$. The linear-in-$\partial\Gamma$ Riemann piece is

$$(\partial\Gamma)^\rho{}_{\sigma\mu\nu} := \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma}. \tag{4}$$

Antisymmetry in $(\mu, \nu)$ is one ring discharge; the full coordinate Riemann is

$$R^\rho{}_{\sigma\mu\nu} = (\partial\Gamma)^\rho{}_{\sigma\mu\nu} + (\Gamma\Gamma)^\rho{}_{\sigma\mu\nu}, \tag{5}$$

inheriting antisymmetry-in-$(\mu, \nu)$ from both summands.

### B. Algebraic first Bianchi

The algebraic first Bianchi identity is $R^\rho{}_{\sigma\mu\nu} + R^\rho{}_{\mu\nu\sigma} + R^\rho{}_{\nu\sigma\mu} = 0$. We discharge it under two torsion-free hypotheses, one each for $\Gamma$ and $\partial\Gamma$.

**Theorem 5** (Linear-piece first Bianchi). *Under `ChristoffelPartialTorsionFree` ($d\Gamma\,\mu\,\rho\,\nu\,\sigma = d\Gamma\,\mu\,\rho\,\sigma\,\nu$ for all $\mu, \rho, \nu, \sigma$), the cyclic sum of* (4) *over $(\sigma, \mu, \nu)$ vanishes.*

The proof `linear_combination`-discharges three torsion-free swap equations weighted with signs $(+, -, -)$, after which the six $\partial\Gamma$ summands cancel pairwise.

**Theorem 6** (Quadratic-piece first Bianchi). *Under `IsTorsionFree` ($\Gamma^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu}$), the cyclic sum of* (2) *over $(\sigma, \mu, \nu)$ vanishes.*

This is the load-bearing new theorem on top of the W7 quadratic content. W7 flagged the quadratic-only first Bianchi as "trivial under bundled commutator construction" — that observation is correct in the bundled formulation $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$ since the cyclic sum is automatically zero by the algebraic identity for nested-commutator cycles. But at the algebraic Christoffel-formula level the cyclic-sum cancellation is genuinely an algebraic identity that requires torsion-freeness explicitly: the proof unfolds twelve $\sum_\lambda$ summands (six $\Gamma\Gamma$ products each appearing in two cyclic positions), rewrites the inner Christoffel index pairs via three applications of `IsTorsionFree` per $\lambda$, and closes with `ring`.

**Theorem 7** (Full-Riemann algebraic first Bianchi). *Under `IsTorsionFree` on $\Gamma$ and `ChristoffelPartialTorsionFree` on $d\Gamma$, the cyclic sum of* (5) *over $(\sigma, \mu, \nu)$ vanishes.*

This is the wave-headline classical-GR algebraic identity. By $\Gamma\Gamma + (\partial\Gamma)$ linearity, it follows from 5 and 6. The result lifts to the cyclic-sum identity on the lowered $(0, 4)$-tensor via SK-EFT-Hawking's `Curvature.firstBianchi_lift` and combines with the metric-compatible-connection antisymmetry-in-first-pair to yield $R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}$ via `Curvature.pair_symmetry_lowered`.

### C. Multi-session plan

This module's content is Session 1 of a five-session plan to a bundle-level Mathlib-PR-shaped Riemann curvature on the Bonn `CovariantDerivative` API. Session 2 adds $\partial^2\Gamma$ + Schwarz symmetry + the linear-piece differential second Bianchi $\nabla_\lambda R + \nabla_\mu R + \nabla_\nu R = 0$ cyclically (which mechanically unblocks the $\nabla^\mu G_{\mu\nu} = 0$ identity in `EinsteinTensor.lean` once the Session 4c specialisation lands). Session 3 ships `BundleRiemannAux.lean` consuming `IsCovariantDerivativeOn` + `VectorField.mlieBracket`. Sessions 4a + 4b ship `BundleRiemann.lean` with cyclic Jacobi for mlieBracket + the abstract bundle-level first Bianchi + root-namespace `IsCovariantDerivativeOn.neg` / `.sub` Bonn-API helpers. Sessions 5 + 5b ship `LeviCivita.lean` with the substantive bundle-level Levi-Civita uniqueness via the Koszul-bilinear-form derivation. Session 4c (deferred Mathlib-PR-track infrastructure) builds `TensorialAt.mkHom`$_3$ from existing `mkHom`$_2$ for the three-argument $R(X, Y, Z)$ and the specialisation to coordinate scope.

## VII. DIFFERENTIAL SECOND BIANCHI MACHINERY

**Session 2 of the Wave 8 plan** adds the second-derivative carrier $\partial^2\Gamma$, the covariant-derivative operator on a generic Riemann tensor, and the substantive coordinate-level discharges that are tractable at this scope. The full coordinate-level differential second Bianchi for the entire Riemann tensor (cubic-$\Gamma$ pieces from $\partial(\Gamma\Gamma) \times \Gamma$) is deferred — at bundle scope (Session 3) it falls out from the Jacobi identity of the connection commutator in three lines, which the Session 4 specialisation theorem then delivers back at coordinate level.

### A. $\partial^2\Gamma$ carrier and Schwarz hypothesis

The covariant derivative of a Riemann tensor needs the second partial of the Christoffel symbols, $\partial_\lambda \partial_\mu \Gamma^\rho{}_{\nu\sigma}$. We model it as $d^2\Gamma : \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$ with the convention $d^2\Gamma\,\lambda\,\mu\,\rho\,\nu\,\sigma = \partial_\lambda \partial_\mu \Gamma^\rho{}_{\nu\sigma}$. The Schwarz / Clairaut equality of mixed second partials,

$$\partial_\lambda \partial_\mu \Gamma^\rho{}_{\nu\sigma} = \partial_\mu \partial_\lambda \Gamma^\rho{}_{\nu\sigma}, \tag{6}$$

holds for $C^2$ Christoffel fields. At our algebraic-precedent scope (6) is encoded as the predicate `IsSchwarz`; in the Session 3 upstream port this is replaced by `mfderiv` acting on a $C^2$ connection 1-form on the manifold.

### B. Covariant derivative on a generic Riemann tensor

For a (1,3)-tensor $T^\rho{}_{\sigma\mu\nu}$ the covariant derivative is

$$\nabla_\lambda T^\rho{}_{\sigma\mu\nu} = \partial_\lambda T^\rho{}_{\sigma\mu\nu} + \Gamma^\rho{}_{\lambda\alpha} T^\alpha{}_{\sigma\mu\nu} - \Gamma^\alpha{}_{\lambda\sigma} T^\rho{}_{\alpha\mu\nu} - \Gamma^\alpha{}_{\lambda\mu} T^\rho{}_{\sigma\alpha\nu} - \Gamma^\alpha{}_{\lambda\nu} T^\rho{}_{\sigma\mu\alpha}. \tag{7}$$

We implement this generically on a `RiemannTensor` together with its partial $\partial T$ : `RiemannPartial` as the operator `covRiemann`. The data-driven specialisation `covRiemann_coord` feeds the partial computed from $(\Gamma, d\Gamma, d^2\Gamma)$ via the product rule.

The $(\mu, \nu)$-antisymmetry of $T$ lifts through the cov derivative (7): each of the four $\Gamma \times R$ correction terms preserves antisymmetry pairwise, the lower-index $\Gamma$-correction pair $-\Gamma^\alpha{}_{\lambda\mu} R^\rho{}_{\sigma\alpha\nu} - \Gamma^\alpha{}_{\lambda\nu} R^\rho{}_{\sigma\mu\alpha}$ swapping into itself with the appropriate sign under $\mu \leftrightarrow \nu$.

### C. Linear-piece differential second Bianchi (wave headline)

**Theorem 8** (Linear-piece differential second Bianchi). *Under `IsSchwarz` on $d^2\Gamma$, the cyclic sum of the partial-of-linear-Riemann-piece over $(\lambda, \mu, \nu)$ vanishes:*

$$\partial_\lambda\big(\partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma}\big) + \partial_\mu\big(\partial_\nu \Gamma^\rho{}_{\lambda\sigma} - \partial_\lambda \Gamma^\rho{}_{\nu\sigma}\big) + \partial_\nu\big(\partial_\lambda \Gamma^\rho{}_{\mu\sigma} - \partial_\mu \Gamma^\rho{}_{\lambda\sigma}\big) = 0. \tag{8}$$

This is the substantive Schwarz-only discharge. Three pairs of $d^2\Gamma$ summands cancel exactly under $d^2\Gamma_{\lambda\mu} = d^2\Gamma_{\mu\lambda}$, $d^2\Gamma_{\mu\nu} = d^2\Gamma_{\nu\mu}$, and $d^2\Gamma_{\nu\lambda} = d^2\Gamma_{\lambda\nu}$. No torsion-freeness on $\Gamma$ or first Bianchi on $R$ is needed for the linear piece — it is the only sub-piece of the differential Bianchi that decouples cleanly at coordinate scope.

### D. Group C+D quadratic cancellation

The $\Gamma \times R$ correction terms in (7) split into four groups by which index of $R$ the contraction-with-$\Gamma$ index $\alpha$ lands on. Under the cyclic sum in $(\lambda, \mu, \nu)$, the lower-index pair of groups (the $\mu$- and $\nu$-position contractions, denoted Groups C and D) cancels by torsion-freeness on $\Gamma$ plus `AntisymLastTwo` on $R$ alone:

**Theorem 9** (Group C+D cyclic cancellation). *For any $\Gamma$ with `IsTorsionFree` and any $R$ with `AntisymLastTwo`,*

$$\sum_{\text{cyclic }(\lambda,\mu,\nu)} \Big( -\sum_\alpha \Gamma^\alpha{}_{\lambda\mu} R^\rho{}_{\sigma\alpha\nu} - \sum_\alpha \Gamma^\alpha{}_{\lambda\nu} R^\rho{}_{\sigma\mu\alpha} \Big) = 0. \tag{9}$$

The proof is a 48-monomial signed-pair cancellation: per-$\alpha$, $\Gamma^\alpha{}_{\lambda\mu} = \Gamma^\alpha{}_{\mu\lambda}$ (torsion-free) and $R^\rho{}_{\sigma\alpha\nu} = -R^\rho{}_{\sigma\nu\alpha}$ (antisym), giving the inter-cyclic match across all four index values $\alpha \in \{0, 1, 2, 3\}$ in each of the six summands. This is the cleanest non-trivial chunk of the full differential Bianchi that holds at coordinate scope.

### E. Strategic deferral of the full discharge to Session 4

The cyclic sum of $\nabla_\lambda R^\rho{}_{\sigma\mu\nu}$ over $(\lambda, \mu, \nu)$ decomposes into four pieces, each requiring distinct hypothesis combinations:

- $(\partial^2\Gamma)$-piece: cancels under `IsSchwarz` alone (Theorem 8).
- $(\partial\Gamma \times \Gamma)$ bilinear piece: needs cross-pair matching between $\partial(\Gamma\Gamma)$ and Group A/B $(\Gamma \times \partial\Gamma)$ terms; cancellation requires algebraic first Bianchi on $R$ + torsion-freeness on $\Gamma$.
- $(\Gamma\Gamma\Gamma)$ cubic piece: cancels under torsion-free + algebraic first Bianchi via standard symmetric-pair matching.
- Group C+D lower-index correction: cancels under torsion-free + `AntisymLastTwo` alone (Theorem 9).

The bilinear and cubic parts are computationally tractable at coordinate scope but produce hundreds of `ring`-normalised monomials. At bundle scope (Wave 8 Session 3, the Bonn `IsCovariantDerivativeOn` consumer; see §VIII below for the Session 3 module that opens that scope), the same content discharges in three lines via `mlieBracket_jacobi`. The strategic ordering of the five-session plan trades a one-session delay for a ten-fold simpler proof, and the Session 4 specialisation theorem then delivers the full coordinate-level identity as a consequence.

## VIII. BUNDLE-LEVEL RIEMANN CURVATURE ON THE BONN API

The bridge from coordinate-scope (Sessions 1+2) to bundle-scope (Sessions 3–5) is opened by the Session 3 module `BundleRiemannAux.lean`, which lifts the curvature operator from the algebraic-precedent $\mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathrm{Fin}\,4 \to \mathbb{R}$ data array to a function $(\mathrm{cov}, X, Y, Z, x) \mapsto V\,x$ on a vector bundle $V \to M$ over a manifold $M$ over a nontrivially-normed field $\mathbb{K}$. The curvature operator is defined verbatim from Klingenberg's textbook formulation [9],

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z, \tag{10}$$

unrolled into the Bonn-API argument-order convention $\mathrm{cov}\,\sigma\,x\,(X\,x) \equiv \nabla_X\,\sigma\,x$:

$$\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,Y\,Z\,x \;=\; \mathrm{cov}(\,y \mapsto \mathrm{cov}\,Z\,y\,(Y\,y)\,)\,x\,(X\,x) - \mathrm{cov}(\,y \mapsto \mathrm{cov}\,Z\,y\,(X\,y)\,)\,x\,(Y\,x) - \mathrm{cov}\,Z\,x\,(\mathrm{mlieBracket}\,I\,X\,Y\,x). \tag{11}$$

### A. Wave-headlines: $R(X, X)Z = 0$ and antisymmetry-$(X, Y)$

The two wave-headlines of Session 3 are purely algebraic consequences of (a) the manifold Lie bracket lemmas `VectorField.mlieBracket_self` and `VectorField.mlieBracket_swap_apply` in [10], and (b) the type-level continuous-linearity of $\mathrm{cov}\,\sigma\,x : T_xM \to_L V_x$ (`map_zero` and `map_neg`). They do *not* require cov to be a covariant derivative.

**Theorem 10** ($R(X, X)Z = 0$). *For any* cov : $(\Pi\,x : M, V\,x) \to (\Pi\,x : M, T_xM \to_L V_x)$, *any vector field* $X$, *any section* $Z$, *and any point* $x$, $\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,X\,Z\,x = 0$.

The proof unfolds the definition; the first two summands cancel immediately ($A - A = 0$); the third summand vanishes via `mlieBracket_self` : $\mathrm{mlieBracket}\,I\,X\,X = 0$ (at point $x$, $0 : T_xM$) followed by $(\mathrm{cov}\,Z\,x)$.`map_zero`.

**Theorem 11** (Antisymmetry-$(X, Y)$ at a point). *For any* cov*, any vector fields* $X, Y$*, any section* $Z$*, and any point* $x$*,* $\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,Y\,Z\,x = -\,\texttt{bundleRiemannAux}\,\mathrm{cov}\,Y\,X\,Z\,x$.

The proof rewrites the bracket via `mlieBracket_swap_apply` : $\mathrm{mlieBracket}\,I\,X\,Y\,x = -\,\mathrm{mlieBracket}\,I\,Y\,X\,x$, then linearity of $\mathrm{cov}\,Z\,x$ converts $\mathrm{cov}\,Z\,x\,(-v) = -\,\mathrm{cov}\,Z\,x\,v$, and the four leading summands swap and negate to match. The function-equality companion `bundleRiemannAux_swap` is shipped via `funext` in the Mathlib idiom of pairing `_apply` with its section-level form.

### B. Commuting-vector-fields specialisation

When the Lie bracket vanishes at a point (e.g. when $X$ and $Y$ are coordinate vector fields, or commute by some structural reason), the curvature reduces to the pure connection commutator. We ship this as a substantive named simplification:

**Theorem 12** (Commuting-vector-fields specialisation). *Under the hypothesis* $\mathrm{mlieBracket}\,I\,X\,Y\,x = 0$, $\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,Y\,Z\,x = \mathrm{cov}(\,y \mapsto \mathrm{cov}\,Z\,y\,(Y\,y)\,)\,x\,(X\,x) - \mathrm{cov}(\,y \mapsto \mathrm{cov}\,Z\,y\,(X\,y)\,)\,x\,(Y\,x)$.

This is the form encountered in textbook coordinate calculations (where $X = \partial_\mu, Y = \partial_\nu$ have vanishing bracket) and is the structural cross-bridge to the Sessions-1+2 coordinate Riemann that Session 4's specialisation theorem will upgrade.

### C. Zero-degeneracies on Bonn's `IsCovariantDerivativeOn`

Two further structural identities ship under the hypothesis $\texttt{IsCovariantDerivativeOn}\,F\,\mathrm{cov}\,\mathrm{Set.univ}$ and the typeclass $[\texttt{VectorBundle}\,\mathbb{K}\,F\,V]$, which together supply $\mathrm{cov}\,0 = 0$ via Bonn's `IsCovariantDerivativeOn.zero` [1].

**Theorem 13** ($R(X, Y)\,0 = 0$). $\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,Y\,(0 : \Pi\,x, V\,x)\,x = 0$, *with the proof composing* $\mathrm{cov}\,0 = 0$ *at the inner sections* $(\,y \mapsto \mathrm{cov}\,0\,y\,(Y\,y)\,) = 0$ *and* $(\,y \mapsto \mathrm{cov}\,0\,y\,(X\,y)\,) = 0$, *and at the outer* $\mathrm{cov}\,0\,x = 0$.

**Theorem 14** ($R(0, Y)\,Z = 0$). $\texttt{bundleRiemannAux}\,\mathrm{cov}\,(0 : \Pi\,x, T_xM)\,Y\,Z\,x = 0$, *composing four distinct Mathlib lemmas:* `Pi.zero_apply` *for the outer X-slot;* $(\mathrm{cov}\,\sigma\,x)$.`map_zero`*;* $(\,y \mapsto \mathrm{cov}\,Z\,y\,0\,) = 0$ *followed by* `IsCovariantDerivativeOn.zero`*;* `mlieBracket_zero_left` *for the bracket term.*

The dual $R(X, 0)\,Z = 0$ falls out as a one-line corollary of Theorem 11 composed with Theorem 14 and is intentionally not shipped to avoid P3-trivial corollary plumbing.

### D. Why bundle scope (and what is intentionally deferred to Session 4)

The Bianchi identities at bundle scope follow with much less arithmetic than at coordinate scope because they reduce to algebraic properties of the Lie bracket. The algebraic first Bianchi at bundle scope for torsion-free cov, $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$, follows from torsion vanishing plus the Jacobi identity for mlieBracket in three lines. The differential second Bianchi at bundle scope, $(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0$, falls out from Jacobi for mlieBracket and from $(\nabla_X R)$ being itself a tensor, also in three lines.

These payoffs make the bundle-level Bianchi the right architecture: at coordinate scope the same identities require $\sim 500$ monomials of `ring`-discharge after dozens of explicit rewrites (per Sessions 1+2 strategic deferrals). Sessions 4a + 4b ship the abstract bundle-level first Bianchi as a $\sim$ 50-line proof closing via cyclic Jacobi for mlieBracket and three torsion-free regroupings; the Bonn-API-discharged form lands in §9 below. Session 4c infrastructure ($\texttt{TensorialAt.mkHom}_3$ build + bundled `CovariantDerivative.riemann` + tensoriality + specialisation to coordinate scope on

EuclideanSpace $\mathbb{R}$ (Fin 4) delivering Sessions-1+2 identities together with the load-bearing $\nabla^\mu G_{\mu\nu} = 0$ discharge for `EinsteinTensor.lean`) is deferred as a Mathlib-PR-track follow-on; the Session 4b Bonn-wrapped Bianchi is already directly applicable from `IsCovariantDerivativeOn` at consumer sites.

## IX. BUNDLE-LEVEL CYCLIC JACOBI AND ABSTRACT FIRST BIANCHI (SESSIONS 4A + 4B)

`BundleRiemann.lean` extends the Session 3 bare-function `bundleRiemannAux` with the algebraic first Bianchi identity under torsion-freeness, when the bundle is the tangent bundle (so the cyclic permutation $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y$ is well-typed). Two layers ship in this module: the abstract layer (Session 4a) parameterised by abstract torsion-free + section-additivity hypotheses, and the Bonn-API-discharged layer (Session 4b) consuming `IsCovariantDerivativeOn` directly.

### A. Pointwise cyclic Jacobi for mlieBracket (Session 4a)

The mathematical core of the bundle-level first Bianchi is the cyclic Jacobi identity

$$[X,[Y,Z]]_x + [Y,[Z,X]]_x + [Z,[X,Y]]_x \;=\; 0$$

for the manifold Lie bracket. Mathlib4 at `81a5d257` ships only the Leibniz form, $[U,[V,W]]_x = [[U,V],W]_x + [V,[U,W]]_x$ (`leibniz_identity_mlieBracket_apply`). Cyclic Jacobi follows from one Leibniz instance plus three antisymmetry rewrites: outer swap $[[X,Y],Z]_x = -[Z,[X,Y]]_x$ via `mlieBracket_swap_apply`; inner swap $[X,Z] = -[Z,X]$ via section-level `mlieBracket_swap`; and inner-negation distribution $[Y,-[Z,X]]_x = -[Y,[Z,X]]_x$ via the new `mlieBracket_neg_right_apply` (which Mathlib also lacks at `81a5d257`). The negation-distribution lemma is itself derived in $\sim$ 5 lines from `mlieBracket_const_smul_right` at $c = (-1 : \mathbb{K})$ plus `neg_one_smul`. The slot-1 sibling `mlieBracket_neg_left_apply` ships in parallel as a Mathlib-PR-shaped helper.

To our knowledge no proof assistant has previously formalised cyclic Jacobi for the manifold Lie bracket; Mathlib's Leibniz form is the closest precedent.

### B. Abstract bundle first Bianchi (Session 4a)

For cov a covariant derivative on the tangent bundle that is torsion-free in the sense of Bonn ($\mathrm{cov}\,W\,x\,(V\,x) - \mathrm{cov}\,V\,x\,(W\,x) = [V,W]_x$ for all sections $V, W$ at every point $x$), and additive in the section argument, the cyclic Bianchi sum vanishes:

$$\texttt{bundleRiemannAux}\,\mathrm{cov}\,X\,Y\,Z\,x + \texttt{bundleRiemannAux}\,\mathrm{cov}\,Y\,Z\,X\,x + \texttt{b}$$

Three torsion-free regroupings (one per axis) collapse the inner double-derivative pairs into bracket-of-brackets terms; the cyclic sum then reduces to cyclic Jacobi. The proof closes via `linear_combination` (`norm := abel`) of the three group equalities plus the cyclic-Jacobi instance. The Christoffel-style cancellation visible at coordinate scope is absorbed into the type-level continuous linearity of $\mathrm{cov}\,\sigma\,x : T_xM \to_L V_x$ and the Bonn-API shape of torsion-freeness.

### C. Root-namespace IsCovariantDerivativeOn helpers (Session 4b)

The abstract first Bianchi requires section-additivity of the connection at the relevant evaluation points, which Bonn's `IsCovariantDerivativeOn.add` supplies directly. To handle the $\mathrm{cov}(\sigma - \tau)\,x = \mathrm{cov}\,\sigma\,x - \mathrm{cov}\,\tau\,x$ form needed by the Bianchi proof, we ship two new root-namespace lemmas in Mathlib upstream-style (mirroring `_root_.ContMDiffAt.mlieBracket_vectorField`): `_root_.IsCovariantDerivativeOn.neg` (derived from Bonn's `.smul_const` at $a = -1$ plus `neg_one_smul`), and `_root_.IsCovariantDerivativeOn.sub` (derived from `.add` + `.neg` + section-level `mdifferentiableAt_neg_section`).

### D. Bonn-API-discharged bundle first Bianchi (Session 4b)

The wave-headline Session 4b deliverable

`bundleRiemannAux_first_bianchi_torsionfree_of_isCovariantD`

re-expresses the abstract first Bianchi by consuming Bonn's `IsCovariantDerivativeOn` directly. The universally-quantified abstract section-additivity hypothesis is replaced by per-pair pointwise differentiability hypotheses on the six section pairs ($\mathrm{cov}\,\cdot\,(\cdot)$) that the proof actually uses; each is discharged via hcov.sub. This gives consumers an interface that depends only on the connection being a Bonn covariant derivative plus pointwise smoothness of the test sections — no abstract additivity plumbing at the consumer site.

### E. Strengthening cuts (Session 4a + 4b ruthless audit)

Three section-level `ext`/`funext` lifts of the pointwise lemmas above were retroactively cut at the post-wave audit as P3 plumbing: `mlieBracket_neg_right` / `mlieBracket_neg_left` / `mlieBracket_cyclic_jacobi` each had no internal or external consumer in the SK-EFT project, with substantive content living in the corresponding _apply versions. Per the project's audit discipline, when there is no immediate consumer and the substantive content is reachable through another lemma, the corollary is cut and re-added at Mathlib-PR-submission time.

## X. BUNDLE-LEVEL LEVI-CIVITA UNIQUENESS VIA THE KOSZUL BILINEAR-FORM ARGUMENT (SESSIONS 5 + 5B)

`LeviCivita.lean` ships the substantive bundle-level Levi-Civita uniqueness theorem (Wald §3.1 / Kobayashi-Nomizu Thm IV.2.2): every torsion-free metric-compatible connection on the tangent bundle of a Riemannian (or non-degenerate Lorentzian) manifold agrees pointwise with every other such connection, and the equality follows from the Koszul-bilinear-form identity rather than from a predicate-level def-as-equality. Two layers ship: the algebraic uniqueness kernel (Session 5) and the substantive `IsLeviCivita` predicate plus Koszul-derivation + composition-uniqueness theorem (Session 5b).

### A. Algebraic uniqueness kernel (Session 5)

The kernel theorem

`leviCivita_pointwise_unique_of_koszul`

takes two abstract connections $\mathrm{cov}, \mathrm{cov}'$ on the tangent bundle and a non-degenerate bilinear form $g$, plus the hypothesis that for every test section triple $(X, Y, Z)$ at the chosen point $x$,

$$g_x(\mathrm{cov}\, Y\, x\, (X\, x), Z\, x) = g_x(\mathrm{cov}'\, Y\, x\, (X\, x), Z\, x).$$

The conclusion is that $\mathrm{cov}\, Y\, x\, (X\, x) = \mathrm{cov}'\, Y\, x\, (X\, x)$ pointwise. The proof routes via bilinearity of $g_x$ ($g_x(v, w) = 0$ for all $w$ implies $v = 0$ at non-degenerate $g_x$) plus `FiberBundle.extend` $E\, w$ to realise an arbitrary tangent vector $w \in T_x M$ as the value at $x$ of some section $Z$. The non-degeneracy hypothesis is encoded as the auxiliary def `IsNondegenerateAt` $g\, x$.

### B. Substantive IsLeviCivita predicate (Session 5b)

The Levi-Civita predicate ships as a four-conjunct `structure` on $(\mathrm{cov}, g)$:

- is_cov: cov is a covariant derivative (`IsCovariantDerivativeOn` $E$ cov Set.univ).
- torsion_free: $\mathrm{cov}\, W\, x\, (V\, x) - \mathrm{cov}\, V\, x\, (W\, x) = [V, W]_x$ for all section pairs.
- metric_symm: $g_x(u, v) = g_x(v, u)$.
- metric_compatible: under sufficient smoothness of the test sections and the scalar pairing $y \mapsto g_y(Y\, y, Z\, y)$,

$$\mathrm{extDerivFun}\, (y \mapsto g_y(Y\, y, Z\, y))\, x\, (X\, x) = g_x(\mathrm{cov}\, Y\, x\, (X\, x), Z\, x) + g_x($$

The metric-compat condition uses Mathlib's extDerivFun (clean $\mathbb{K}$-valued exterior derivative wrapper) rather than mfderiv directly, which avoids the $\mathrm{TangentSpace}\, \mathcal{I}(\mathbb{K}, \mathbb{K})\, v$ type-elaboration friction at small cost in surface area.

An in-session reflective strengthening cut earlier in Session 5 removed an initial three-conjunct draft whose metric_symm conjunct was the only one constraining the cov-$g$ relationship; the substantive metric-compat conjunct was added in 5b.

### C. Koszul identity from IsLeviCivita (Wald §3.1 derivation)

The wave-headline theorem

`koszul_identity_of_isLeviCivita`

derives the Koszul bilinear-form identity from `IsLeviCivita`:

$$\begin{aligned} 2 \cdot g_x(\mathrm{cov}\, Y\, x\, (X\, x), Z\, x) \;=\; & \mathrm{extDerivFun}\, (g(Y, Z))\, x\, (X\, x) \\ & + \mathrm{extDerivFun}\, (g(Z, X))\, x\, (Y\, x) \\ & - \mathrm{extDerivFun}\, (g(X, Y))\, x\, (Z\, x) \\ & + g_x([X, Y]_x, Z\, x) - g_x([X, Z]_x, Y\, x) - \end{aligned}$$

The proof applies the metric-compat conjunct at three cyclic shifts $(X, Y, Z)$, $(Y, Z, X)$, $(Z, X, Y)$; rewrites the slot-2 connection terms to slot-1 via three metric-symmetry instances; and substitutes torsion-free at three cross-pairs to fold the $g(\nabla \cdot \cdot, \cdot) - g(\nabla \cdot \cdot, \cdot)$ differences into bracket pairings. The combined step closes via `linear_combination` with derived coefficients $[-1, -1, +1, -1, -1, +1, -1, -1, +1]$ across the nine hypotheses (three metric-compat, three symmetry rewrites, three torsion-free substitutions). The right-hand side depends only on $g$, $X$, $Y$, $Z$ (not on cov), so any two `IsLeviCivita` connections produce equal $g$-pairings of $\mathrm{cov}\, Y\, x\, (X\, x)$ against $Z\, x$.

### D. Composition uniqueness theorem (Session 5b)

Composing the Koszul identity (Session 5b) with the algebraic uniqueness kernel (Session 5) gives the full uniqueness:

`leviCivita_unique_of_isLeviCivita`.

For any two `IsLeviCivita` connections cov and $\mathrm{cov}'$ of the same metric $g$, at any non-degenerate point $x$,

$\mathrm{cov}\, Y\, x\, (X\, x) = \mathrm{cov}'\, Y\, x\, (X\, x)$. The $(2 : \mathbb{K}) \neq 0$ hypothesis is unavoidable: the Koszul derivation yields a factor of 2 on the left-hand side, so two connections satisfying the same Koszul identity differ by an element annihilated by 2, which is 0 iff $\mathrm{char}(\mathbb{K}) \neq 2$. The cancellation step uses `mul_left_cancel₀`.

This is the substantive Wald §3.1 / Kobayashi-Nomizu IV.2.2 uniqueness theorem at bundle level, deferred from `RiemannianConnection.lean`'s coordinate-scope predicate-level form (which was a P5 trivial-trans on the def-as-equality $\Gamma =$ `christoffelKoszul` and was retroactively cut at the W7+W8(S1-3) reflexive strengthening pass). To our knowledge no proof assistant has previously formalised bundle-level Levi-Civita uniqueness via the substantive Koszul-bilinear-form derivation.

### E. Strengthening cut (Session 5 + 5b ruthless audit)

The section-level `leviCivita_unique` theorem (`FiberBundle.extend` lift of the pointwise kernel from "section $X$ with $X\,x = w$" to "arbitrary tangent vector $V \in T_xM$") was retroactively cut at the post-wave audit as P3 plumbing. Substantive content lives in `leviCivita_pointwise_unique_of_koszul` (the kernel, X-form) and in `leviCivita_unique_of_isLeviCivita` (the composition theorem, which routes through the X-form). The V-form lift had no internal/external consumer; per audit discipline, cut and re-add at Mathlib-PR-submission time if needed.

## XI. WHAT THIS UNBLOCKS

The seven new modules are infrastructure: they ship no new physics and no new submission-ready prediction. Their value is in unblocking five deferred items in the Phase 6f programme that were previously parked behind the Mathlib indefinite-signature gap and the Riemann-from-connection gap. The W8 Session 1 first-Bianchi discharge in particular promotes the unblock from "mechanical pending Mathlib" to "mechanical pending an internal SK-EFT module" for several items:

- **6f.2 second Bianchi.** $\nabla^\mu G_{\mu\nu} = 0$ requires a covariant-divergence operator on the Einstein tensor. With `christoffelKoszul` in hand and the algebraic first Bianchi shipped, the divergence can be expressed as an algebraic identity in $(g, \partial g, \partial^2 g)$ and the second Bianchi proven directly. The W8 Session 2 module `RiemannDifferentialBianchi.lean` ships the $\partial^2\Gamma$ carrier with the Schwarz hypothesis, the covariant-derivative operator on a generic Riemann tensor, the linear-piece differential second Bianchi under Schwarz alone, and the Group C+D quadratic cyclic cancellation under torsion-freeness plus `AntisymLastTwo`. The full coordinate-level differential second Bianchi for the entire Riemann tensor is deferred to the W8 Session 4 specialisation theorem, which delivers it as a consequence of the Session 3 bundle-level Jacobi identity. The remaining gap between the formalisation and `EinsteinTensor.lean`'s $\nabla^\mu G_{\mu\nu} = 0$ theorem statement is therefore the Session 4 specialisation; both layers (bilinear and cubic) of the full discharge are themselves three-line bundle-level statements.
- **6f.4 Schwarzschild vacuum-Ricci.** The algebraic-precedent statement $R_{\mu\nu} = 0$ for the Schwarzschild solution was rejected at first pass per the 6f.2 lesson (the tracked-Prop form was vacuous because we had no Christoffel-side machinery). With the full coordinate Riemann including the linear-in-$\partial\Gamma$ piece shipped, the $R^\rho{}_{\sigma\rho\nu} = 0$ contraction at the algebraic level is now mechanical: instantiate $\Gamma$ and $\partial\Gamma$ to the Schwarzschild values, contract over the upper Riemann index, and discharge by direct computation. This is naturally a follow-on consumer module after Session 2.
- **6f.5 ADM 3D Riemannian Ricci on Cauchy slices.** The $D_j$ covariant-divergence on a 3D Cauchy slice was deferred for the same reason. With the full coordinate Riemann shipped, the 3D restriction is direct: instantiate `Christoffel` to the spatial connection and reuse the same algebraic-precedent pattern.
- **6f.6 Cartan structure equations.** $d\omega + \omega \wedge \omega = R$ requires the connection 1-form $\omega$, which is naturally read off the Christoffel symbols via $\omega^a{}_{b\mu} = e^a{}_\nu \Gamma^\nu{}_{b\mu} - e^a{}_{b,\mu}$. With Christoffel symbols formalized and the algebraic first Bianchi corollary shipped (which gives the algebraic identity $d^2\omega = 0$ in the cyclic-sum form), the bridge is a tetrad/connection-1-form arithmetic exercise once the exterior derivative $d$ is modelled (W8 Session 4 bundle-level layer is the natural place; or a separate `DifferentialFormsCoordinate` sibling module).
- **6g.2 / 6g.6 curve-theoretic Penrose / Mazur.** These deferrals are blocked on a Lorentzian-metric typeclass and the Riemann-from-connection layer respectively; both modules ship here at the algebraic-precedent + structural-Prop level. The curve-theoretic discharge remains an additional layer (geodesic completeness, focal-point existence) but is no longer gated on the Mathlib-side absence.

### A. 2026-04-30 follow-on: Phase 6g.2 substantive curve-theoretic Penrose shipped

The W8 Session 5+5b bundle-level Levi-Civita uniqueness backbone unlocked the substantive curve-theoretic

Penrose chain as the natural Mathlib-PR-shape follow-on. A 5-session multi-session wave shipped this content on 2026-04-30, end-to-end, immediately following this paper's infrastructure delivery. The substantive content lifts as **D3 §24** (the headline correctness-push section, per the project bundle architecture in `docs/PAPER_DRAFT_MAPPING.md`), not as an extension of this paper's I1 sidebar / D3 §22.5 lift, so we do not absorb the curve-theoretic content here. We summarise the follow-on briefly:

- **LorentzianBundle.lean** (5 substantive theorems + 2 structures): bundle-level Lorentzian metric typeclass `IsLorentzianBundle` (symmetric + non-degenerate fiber bilinear form) with the wave-headline bridge `leviCivita_unique_for_lorentzianBundle` composing this paper's § **??** composition theorem `leviCivita_unique_of_isLeviCivita`; bundle-level Riemannian-vs-Lorentzian falsifier `riemannianBundle_not_lorentzianBundle_with_negative_self_pairing`.
- **NullGeodesic.lean** (4 substantive theorems + 1 structure + 1 def): vector-field-level null auto-parallel congruence with substantive metric-compatibility-driven preservation of the null self-pairing along the flow (`autoparallel_self_metric_compat_dir_deriv_zero` consumes § **??**'s `IsLeviCivita.metric_compatible` directly).
- **RaychaudhuriEquation.lean** (5 substantive theorems + 1 structure): the substantive Raychaudhuri focusing inequality $d_k\theta \leq -\theta^2/(n-1)$ at the abstracted-trace + curve-level scope under hypersurface-orthogonal + non-negative shear + NEC-controlled $R_{kk}$, with $n = 4$ specialisation matching the existing 6g.2 §1 Riccati form $d\theta/d\lambda \leq -\theta^2/3$.
- **FocalPoint.lean** (6 substantive theorems + 1 def): focal-configuration predicate `IsFocalConfigurationFor` + Riccati-comparison bridge to the 6g.2 §1 closed-form `riccatiSolution_neg` / `focusingTime_pos`, with substantive cross-bridge `focal_obstructs_extension` into the existing 6g.2 §3 `IsNullGeodesicallyIncomplete` predicate.
- **PenroseSingularityCurveTheoretic.lean** (5 substantive theorems + 1 structure): the wave-completion composition theorem `penrose_singularity_curve_theoretic` producing geodesic-incompleteness witness + strict-negativity-throughout-window from the 3-conjunct hypothesis bundle `IsCurveTheoreticPenroseHypothesis`.

*a. Wave totals.* 5 new modules / 25 substantive theorems / 5 structures / 2 defs / ∼1909 LOC / 0 sorry / 0 new axioms / library 8485 jobs PASS clean. 2-cut wave-end ruthless audit (in `LorentzianBundle.lean` only: `lorentzianBundle_symm_apply` P7 lift-without-consumer + `riemannianBundle_implies_lorentzian_pairing_pos` P3 trivial-projection-as-named-API; Sessions 2–5 maintained the 0-retroactive-cut streak). The wave ships at *two parallel substantive scopes* — manifold-level abstract focusing inequality (Session 3) and curve-level Riccati-comparison + focal-point bridge (Sessions 4–5); the curve-pullback bridge between these scopes is parked as a Mathlib-PR follow-on and discharges via Mathlib's `Analysis.ODE.Gronwall`.

## XII. CONVENTIONS AND CONFORMANCE

The seven new Lean modules follow Mathlib upstream conventions: file header copyright + module imports + `public` sections, no `maxHeartbeats` overrides, no `ring` on noncommutative contexts. Naming follows Mathlib snake-case for theorems and PascalCase for predicates and structures. The Session 3 module `BundleRiemannAux.lean` additionally adopts Bonn's variable section verbatim (model-with-corners $I$, charted-space $M$, vector bundle $V \to M$ with model fiber $F$), uses `omit` clauses on the algebraic theorems that do not consume $[\forall x, \mathrm{IsTopologicalAddGroup}\,(V\,x)]$ / $[\forall x, \mathrm{ContinuousSMul}\,\mathbb{K}\,(V\,x)]$ in the Mathlib upstream-style idiom (cf. `Mathlib.Geometry.Manifold.-ContMDiff.Atlas` for parallel `omit` usage), and opens `VectorField` so that `mlieBracket_self` and `mlieBracket_swap_apply` resolve directly without fully-qualified names. Cross-module bridges from the `LorentzianMetric` module reach into existing `LinearizedEFE.`$\eta$`_symm`; the `RiemannianConnection` module reaches into `Curvature.AntisymLastTwo`; the `RiemannCoordinate` module reaches into both `Curvature.FirstBianchi` (the wave-headline first-Bianchi predicate consumer) and W7's `RiemannianConnection.IsTorsionFree` + `RiemannianConnection.riemannQuadraticTerm`. All cross-module calls are SK-EFT-internal and survive the eventual upstream port — they will be replaced by their Mathlib-bundle-level counterparts when the `IsContinuousLorentzianBundle` typeclass and the `CovariantDerivative`-consuming Riemann tensor land upstream.

The library passes the project's preemptive-strengthening discipline [5]: each theorem statement is non-vacuous under its hypothesis, no ∃-absorption, no bundle-redundancy, and the Riemannian-Lorentzian falsifier and the metric-partial-symmetry-load-bearing Koszul torsion-freeness preempt P3/P5-trivial restatements.

### A. Strengthening cuts

*a. W7 first-pass cut.* The W7 wave applied a single retroactive cut at first-pass review: `zeroChristoffel_isTorsionFree`, a pure-`rfl` P3-trivial witness with no downstream consumer (zero-symmetry follows by definition without consuming any hypothesis).

*b. W8 Session-1+2+3 first-pass cuts.* W8 Sessions 1, 2, and 3 each shipped with zero retroactive cuts at first pass and after the ruthless audit — matching the project best-pass discipline metric (6e.5 / 6b.2 / 6f.6 baselines).

*c. W8 post-Session-3 strengthening pass (two retroactive cuts).* A reflexive end-of-session strengthening pass after Session 3 found two cut candidates inherited from W7 that the W7 first-pass review missed:

1. `IsLeviCivitaForKoszul` predicate def + `leviCivita_unique_at_koszul` theorem, both in `RiemannianConnection.lean`. The predicate `IsLeviCivitaForKoszul` $\Gamma\, g^{-1}\, \partial g$ := $\Gamma$ = `christoffelKoszul` $g^{-1}\, \partial g$ is literally the equality "$\Gamma$ equals the closed-form Koszul Christoffel". The "uniqueness" theorem proof was `rw [h, h']` — pure transitivity of equality through definitional unfolding. This is a P5 trivial-trans on a def-as-equality predicate. The substantive Levi-Civita uniqueness — the Wald §3.1 / Kobayashi-Nomizu Thm IV.2.2 fundamental theorem that every torsion-free metric-compatible connection equals the Koszul Christoffel — needs the bundle-level Koszul-bilinear-form argument over arbitrary smooth vector fields and is the Wave 8 Session 5 deliverable. Documented in Section IV Remark IV 0 a.

2. `lorentzian_metric_nonzero` theorem in `LorentzianMetric.lean`. The conclusion "$g\,0\,0 \neq 0$" follows from the `IsLorentzianSignature` predicate's first conjunct "$g\,0\,0 = -1$" by one-step `rw + norm_num` discharge of $-1 \neq 0$. The substantive non-vacuity content lives in the `IsLorentzianSignature` predicate definition itself; the determinant-discriminator content lives in `lorentzian_diagonal_product_neg_one`. No downstream consumer.

Both cuts retired predicate-level trivial-trans / corollary patterns that had no downstream consumers and whose substantive content lives elsewhere in the library. The pattern lesson is logged in `feedback_post_wave_strengthening_audit.md`: predicate-as- equality definitions *should not* appear with companion "uniqueness" theorems at the algebraic-precedent scope; defer substantive uniqueness to the bundle-level wave where the Koszul bilinear-form argument supplies the load-bearing content.

### B. 2026-04-30 ruthless audit (Sessions 4a / 4b / 5 / 5b) section-level lift cleanup

A second ruthless post-wave audit on the same day cleaned up the W8 Sessions 4a / 4b / 5 / 5b first-pass declarations and applied four additional retroactive cuts. Each cut targets a section-level `ext` / `funext` / `FiberBundle.extend` lift of a pointwise lemma without any internal or external consumer:

1. `mlieBracket_neg_right` (BundleRiemann.lean §1) — 1-line `ext` $x$; `exact ... apply` $(h\,x)$ lift of the pointwise version. The pointwise `mlieBracket_neg_right_apply` is the internal consumer (used by cyclic-Jacobi); the section-level lift had no consumer.
2. `mlieBracket_neg_left` — same pattern, slot-1.
3. `mlieBracket_cyclic_jacobi` (BundleRiemann.lean §2) — 2-line section-level lift of cyclic-Jacobi-pointwise. Internal consumer is the pointwise version (used by both bundle-Bianchi proofs); section-level form had no consumer.
4. `leviCivita_unique` (LeviCivita.lean §3) — `FiberBundle.extend` lift from the pointwise X-form kernel ("section $X$ with $X\,x = w$") to the V-form ("arbitrary tangent vector $V \in T_x M$"). Substantive content lives in `leviCivita_pointwise_unique_of_koszul` (X-form kernel) and `leviCivita_unique_of_isLeviCivita` (composition theorem, which routes through the X-form). V-form lift had no consumer.

After the cuts the library ships **fifty-two theorems and lemmas across seven modules** (counted by line-start `theorem /lemma` grep), zero `sorry`, zero new project axioms. Supporting structural declarations (`IsLorentzianSignature` / `IsLorentzianMetric` / `IsContinuousLorentzianFamily` / `Christoffel` / `IsTorsionFree` / `IsLeviCivita` as `def` / `structure` declarations) ship alongside the named theorems and lemmas but are not counted in the 52 — they are substrate for the counted results. The pattern lesson logged: section-level `funext/extend` lifts of pointwise lemmas without an internal/external consumer *should not ship in SK-EFT*; defer the section-level companions to Mathlib-PR-submission time as a paired API completion.

## XIII. DISCUSSION AND FOLLOW-UP

### A. Bundle-level Mathlib upstream port

The natural upstream-port target is a `Curvature.lean` file under `Mathlib.Geometry.Manifold.VectorBundle.CovariantDerivati` that defines the Riemann curvature on a bundled

`CovariantDerivative` via the connection-commutator function $R(X,Y)Z := \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]}Z$, proves antisymmetry-in-$(X,Y)$ via the `TensorialAt.mkHom`$_2$ machinery already used in `IsCovariantDerivativeOn.torsion`, then derives Ricci, scalar curvature, the algebraic Bianchi identity (under torsion-free hypothesis), and the bundle-level Levi-Civita uniqueness via the Koszul-bilinear-form argument over smooth vector fields. The algebraic-precedent identities formalized here are the substantive inputs that survive the bundle-level upgrade.

A parallel `Lorentzian.lean` sibling of `Mathlib.Topology.VectorBundle.Riemannian` would supply the `IsContinuousLorentzianBundle` typeclass shape; the algebraic-precedent witness here lifts directly.

### B. Comparison with HoTT / HOL Light / Coq prior art

We are unaware of a Levi-Civita Christoffel formalization in any proof assistant. HOL Light's geometric-algebra library provides Riemannian inner-product spaces but no Christoffel/connection. Isabelle's AFP has nothing in this space. Coq's MathComp has neither Riemannian nor Lorentzian metric infrastructure. PhysLean ports physics formulas but has no curvature library. The HALF-ERC audit report's no-prior-art finding therefore stands.

### C. Multi-session plan and open work

The bundle-level upgrade is an explicit five-session plan tracked in the Phase 6f roadmap (Wave 8, sessions 1–5b). Sessions 1, 2, 3, 4a, 4b, 5, and 5b all ship the content reported in this paper. Session 2 adds `RiemannDifferentialBianchi.lean` with $\partial^2\Gamma$ + Schwarz symmetry + the linear-piece differential second Bianchi identity $\nabla_\lambda R + \nabla_\mu R + \nabla_\nu R = 0$, which mechanically discharges the `EinsteinTensor.`$\nabla^\mu G_{\mu\nu} = 0$ deferral once the Session 4c specialisation lands. Session 3 (§VIII) adds `BundleRiemannAux.lean` opening the Bonn-API bundle scope: the bare-function `bundleRiemannAux` consuming Bonn `IsCovariantDerivativeOn` and `VectorField.mlieBracket` (antisymmetric in $(X,Y)$ from mlieBracket antisymmetry, vanishing on the diagonal from `mlieBracket_self`, with the commuting-vector-fields specialisation and the section/X-slot zero-degeneracies on `IsCovariantDerivativeOn.zero` shipped in the same module). Sessions 4a + 4b (§IX) ship `BundleRiemann.lean`: pointwise cyclic Jacobi for mlieBracket, the abstract bundle-level first Bianchi for $V :=$ TangentSpace $I$ (a ∼50-line proof closed via `linear_combination (norm := abel)`), and the Bonn-API-discharged form of the same theorem consuming `IsCovariantDerivativeOn` directly through new `.neg`/`.sub` root-namespace helpers. Sessions 5 + 5b (§X) ship `LeviCivita.lean`: the algebraic uniqueness kernel (`leviCivita_pointwise_unique_of_koszul`), the substantive 4-conjunct `IsLeviCivita` predicate using Mathlib's `extDerivFun`, the wave-headline `koszul_identity_of_isLeviCivita` derivation closed via `linear_combination`, and the composition uniqueness theorem `leviCivita_unique_of_isLeviCivita` via 2-cancellation under char$(\mathbb{K}) \neq 2$.

**Session 4c (deferred).** A Mathlib-PR-track follow-on builds `TensorialAt.mkHom`$_3$ (from existing `mkHom`$_2$), bundles `bundleRiemannAux` into a trilinear-at-each-fiber `CovariantDerivative.riemann` via `mkHom`$_3$, ships the bundle-level differential second Bianchi as a three-line Jacobi consequence, and specialises on `EuclideanSpace` $\mathbb{R}$ (Fin 4) to deliver the coordinate-level differential second Bianchi at top order plus the $\nabla^\mu G_{\mu\nu} = 0$ discharge for `EinsteinTensor.lean`. Session 4c is not blocking the SK-EFT physics pipeline (the Session 4b `IsCovariantDerivativeOn`-wrapped Bianchi is already directly applicable from Bonn's API at consumer sites); it is parked as Mathlib-PR-track infrastructure. Following Session 4c the Bonn-team outreach email per the project's `phase6f1_curvature_architecture_decision.md` script will be sent and the Mathlib pull request submitted. Each session updates the running state checklist in the Phase 6f roadmap and ships a session-specific project memory; the auto-compact-survival contract is documented in the roadmap's Wave 8 "Session-handoff" subsection.

The full curve-theoretic Penrose and Mazur-Bunting-Israel proofs, also blocked, have additional gates beyond the Lorentzian-metric availability (geodesic existence, focal-point existence, $\sigma$-model rigidity); those are phase-level commitments. Within the SK-EFT-Hawking programme they sit alongside the Phase 6g.7 cosmic-censorship backlog.

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[1] P. Massot, M. Rothgang, H. Macbeth. Mathlib4 covariant derivatives library. *Mathlib4 commit 81a5d257*, April 2026.

Mathlib.Geometry.Manifold.VectorBundle.CovariantDerivative.{Basic,Torsion}.

[2] S. Gouëzel. Riemannian vector bundles in Mathlib4. *Mathlib4 commit 81a5d257*, March 2026. Mathlib.Topology.VectorBundle.Riemannian; Mathlib.Geometry.Manifold.VectorBundle.Riemannian.

[3] J. G. Roehm. Phase 6f waves 1–6 module roster: classical-GR algebraic backbone in Lean. *SK-EFT-Hawking project, in preparation*, inventory snapshot 2026-04-29.

[4] J. G. Roehm. Phase 6g waves 1–6: causal structure, singularity hypothesis bundles, area theorem, no-hair theorem in Lean. *SK-EFT-Hawking project, in preparation*, inventory snapshot 2026-04-29.

[5] J. G. Roehm. Wave-execution pipeline and preemptive-strengthening discipline for theorem-driven theoretical-physics workflows. *SK-EFT-Hawking project, in preparation*, docs/WAVE_EXECUTION_PIPELINE.md.

[6] R. M. Wald. *General Relativity*. University of Chicago Press, 1984. §3.1, 3.2.

[7] S. Kobayashi and K. Nomizu. *Foundations of Differential Geometry, Vol. I.* Wiley-Interscience, 1963. Theorem IV.2.2 (Levi-Civita uniqueness).

[8] S. Carroll. *Spacetime and Geometry.* Addison-Wesley, 2004. §3.2 (Koszul formula).

[9] W. Klingenberg. *Riemannian Geometry*, 2nd ed. de Gruyter Studies in Mathematics, vol. 1, 1995. §1.3 (curvature operator on a vector bundle).

[10] S. Gouëzel. Lie brackets of vector fields on manifolds in Mathlib4. *Mathlib4 commit 81a5d257*, 2024–2026. Mathlib.Geometry.Manifold.VectorField.LieBracket (mlieBracket, mlieBracket_self, mlieBracket_swap_apply, mlieBracket_zero_left, leibniz_identity_mlieBracket).